

Req ID	Type	Requirement	Acceptance Criterion	Justification (Link to Standard)
R-TRACK-01	FR	Robot shall maintain lateral tracking error within tolerance during S1 nominal scenario.	$ e_{line} < 0.15$ m for all $t \in [0,75]$ s	ISO 12100 Clause 5.2: Predictable motion requirement for safe operation. ISO 9241-210 Clause 5.2: Understanding the operational context (line following) defines this core functional need.
R-TRACK-02	FR	Robot shall minimize cumulative tracking error over the full S1 scenario.	$IAE < 0.030$ m·s	Performance optimization derived from user need for precision. ISO 9241-210 Clause 5.3: Translating the high-level user goal of "accurate tracking" into a measurable technical target.
R-OBS-01	FR	Robot shall avoid collision with static obstacles during S2 scenario.	Number of obstacle contacts = 0	ISO 12100: Fundamental safety requirement. ISO 9241-210 Clause 5.2: The context includes unexpected obstacles, so the system must handle them.
R-OBS-02	FR	Robot shall complete the S2 scenario after obstacle avoidance manoeuvre.	Scenario completed without manual intervention	Functional continuity - the system must recover. ISO 9241-210 Clause 5.6: The design must be evaluated on its ability to handle complete scenarios, not just isolated functions.
R-SAFE-01	FR	Robot shall limit speed during sensor fault condition.	$v \leq 0.45$ m/s when $fault_flag = 1$	IEC 61508: Safe state requirement. ISO 9241-210 Clause 5.6: Safety is a key usability outcome - the system must behave predictably even in fault conditions.
R-PERF-01	NFR	Robot shall complete S1 scenario within specified time limit.	$t_{final} \leq 75$ s	ISO 9241-210 Clause 5.3: User efficiency requirement - the task should complete in a reasonable time.
R-ENERGY-01	NFR	Controller shall minimize control energy usage.	Energy proxy < 65	Sustainability consideration. ISO 9241-210 Clause 5.2: Part of the broader context includes environmental impact.
R-ROBUST-01	NFR	Controller shall detect and respond to temporary sensor dropout.	Fault detected within 1 control cycle	ISO 9241-210 Clause 5.6: Reliability directly impacts user trust and system usability. A system that fails silently is not user-centered.
R-HMI-01	NFR	HMI shall display real-time telemetry (e_{line} , v , ω , flags) during simulation.	All variables update at ≥ 10 Hz, clearly labeled	ISO 9241-210 Clause 5.3: Direct translation of user need for "clear feedback" into a measurable requirement. Clause 5.2: Understanding the user's need to monitor system state in real-time.